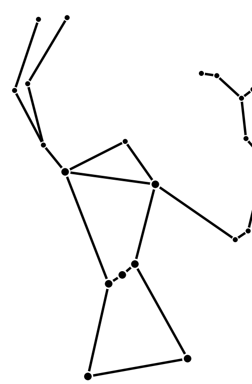
QUOTATIONS

Adobe Caslon Pro 11 (leading 13.2)

github.com/johanley/quotations-ps

QUOTATIONS



Collected by John O'Hanley

1988 – 2025

[Goedel's Theorem] proves that there exists meaningful mathematical statements that are neither provable nor disprovable, now or ever neither provable nor disprovable, that is, not simply because human thought or knowledge is insufficiently advanced but because the very nature of logic renders them incapable of resolution, no matter how long the human race survives or how wise it becomes. The theorem itself was proved some decades ago, and recently the first example of an undecidable mathematical statement was found. Called the Continuum Hypothesis, it is the assertion that in a set-theoretic sense there exists no sets with more elements than the integers but fewer than the set of all real numbers. So there it stands an assertion that to the end of time cannot be proved and yet cannot be disproved. The philosophical implications are devastating.

~ ALFRED ADLER, *Mathematics and Creativity*

Truth emerges more readily from error than from confusion.

~ FRANCIS BACON, *Novum Organum*

Formalism was pioneered by the German mathematician David Hilbert, who developed a grand scheme to turn math into a sort of game where mathematical objects are the playing pieces and axioms are the formal rules of play that determine outcomes.

~ J. S. BARDI, *The Great Math War*

[Brouwer] absolutely refused to admit that anything in mathematics was true or even existed unless it could be explicitly constructed mentally.

~ J. S. BARDI, *The Great Math War*

Other philosophers all came to the same conclusion as Galileo and Aristotle, holding fast to the ancient creed*infinitum actu non datur*(the actual infinite doesn't exist).

~ J. S. BARDI, *The Great Math War*

... and there Russell meets the Italian mathematician Giuseppe Peano, who has developed a logical language the same way Frege had but with better and more accessible symbols.

~ J. S. BARDI, *The Great Math War*

The whole reason Frege developed his*Begriffsschrift*(concept of notation) was that he didn't trust ordinary language.

~ J. S. BARDI, *The Great Math War*

The paradoxes of set theory were the main motivation for wanting to shore up the foundations of math in the first place.

~ J. S. BARDI, *The Great Math War*

But it wasn't until the late nineteenth and early twentieth centuries that *Tertium non datur*began to take on even more significance because mathematicians foundthey could use it along with set theory to treat infinite entities using finitary means. With the law of the excluded middle, you don't need to construct an infinite set in order to work with it.

~ J. S. BARDI, *The Great Math War*

As but one small measure of this glass-ceiling injustice, on June 4, 1919, when she gave her inaugural talk as a teacher (still unpaid at that point), she stood up and described what is now known as Noether's theorem, a follow-up work from her contributions to Hilbert's and Einstein's race for general relativity. Noether's theorem unites symmetry in nature with universal conservation laws, and some consider it a cornerstone not just of general relativity but also of a vast part of modern physics, including particle physics. Some consider it the backbone of upon which all modern physics is built.

~ J. S. BARDI, *The Great Math War*

Goedel says it's possible to create a true formula that claims of itself that it cannot be proven something like*This statement cannot be proven.*Such a statement can be proven only if it is, in fact, itself a lie.

~ J. S. BARDI, *The Great Math War*

Goedel develops a second incompleteness theorem, which says that if a system is inconsistent, it cannot be proven consistent using its own inconsistent means.

~ J. S. BARDI, *The Great Math War*

[Otto Neugebauer] famously declares he's not Jewish by saying, "I did not have the*honor*of having a Jewish grandmother."

~ J. S. BARDI, *The Great Math War*

Contemporary science is going to be proven wrong, but mathematics is not.

~ JOHN D. BARROW, *What is Mathematics?*

The British style eschews general formalism for the sake of elegance alone, is biased towards application, and motivated by the desire to solve practical problems. The French, by contrast, are attracted by formalism and abstraction, as epitomized by the encyclopedic project of the Bourbaki group.

~ JOHN D. BARROW, *What is Mathematics?*

Why is it that abstract mathematical concepts, devised and explored in the distant past with no apparent application, so often turn out to be key elements in the description of some new area of discovery in physics?

~ JOHN D. BARROW, *What is Mathematics?*

Any normal boy or girl of sixteen could master the calculus in half the time often devoted to stumbling through Book One of Caesar's*Gallic War.*

~ E. T. BELL, *The Queen of the Sciences*

The assumptions from which mathematics starts are simple; the rest is not.

~ E. T. BELL, *The Queen of the Sciences*

Mathematics, according to David Hilbert, is a game played according to certain simple rules with meaningless marks on paper.

~ E. T. BELL, *The Queen of the Sciences*

Common sense and all appearances notwithstanding, there are precisely as many real numbers as there are complex numbers.

~ E. T. BELL, *The Queen of the Sciences*

As it is of some importance to understand exactly what Weierstrauss proved, I state it more fully. By retaining*all*the postulates of a field, it is impossible to construct a class of things which satisfies all postulates and which is not either the class of all complex numbers or one of the latter's subclasses.

~ E. T. BELL, *The Queen of the Sciences*

How could the whole race have missed this till Descartes saw it? [Analytic geometry.]

~ E. T. BELL, *The Queen of the Sciences*

All of the astounding inventiveness and infinite variety of geometry during that amazingly prolific half century [1872 1922] is seen as one orderly, simple whole from the commanding summit which Klein recognized as the proper point of view to sweep in the whole of the past of geometry and to foresee much of its future. Here is the famous sentence: "Given a manifold and a group of transformations of the same, to develop the theory of invariants relating to that group."

~ E. T. BELL, *The Queen of the Sciences*

Thus, the description of quantum phenomena requires a distinction in principle between the objects under investigation and the measuring apparatus by means of which the experimental conditions are defined.

~ NIELS BOHR, *Atomic Physics and Human Knowledge*

The very fact that repetition of the same experiment, in general yields different recordings pertaining to the object, immediately implies that a comprehensive account of experience in this field must be expressed by statistical laws.

~ NIELS BOHR, *Atomic Physics and Human Knowledge*

The art of asking questions is more valuable than solving problems.

~ GEORG CANTOR, *Doctoral Thesis*

Yet, the very persistence of such misinterpretations [of SR] must have some deep, underlying cause which a historian of ideas can easily identify: the perennial tradition of both Western and Eastern thought which regards time as merely apparent and not genuinely real.

~ M. CAPEK, *The New Aspects of Time*

The intensity of every astonishment gradually wears off; the human mind, by the sheer effect of repetition and habit, gradually becomes accustomed to even the strangest and least familiar ideas.

~ M. CAPEK, *The Philosophical Impact of Contemporary Physics*

It frequently happens that within one and the same mind the true grasp of the mathematical side of the theory coexists with serious misapprehensions of its physical and especially its philosophical meaning.

~ M. CAPEK, *The Philosophical Impact of Contemporary Physics*

This plainly shows that the term "curvature" is essentially metaphorical, for it is based on an analogy with curvature of two dimensional surfaces.

~ M. CAPEK, *The Philosophical Impact of Contemporary Physics*

...the law of gravitation is a generalized law of inertia.

~ M. CAPEK, *The Philosophical Impact of Contemporary Physics*

What is left of the classical definition of space as "simultaneous juxtaposition of points?" Nothing but a word with misleading connotations.

~ M. CAPEK, *The Philosophical Impact of Contemporary Physics*

Yet the idea of a timeless container, in which all matter is located and in which all motion takes place, has such deep roots in the very structure of human imagination that it is almost impossible to get rid of it.

~ M. CAPEK, *The Philosophical Impact of Contemporary Physics*

...the order and magnitude of the stars and all of their orbits and the heaven itself are so connected that in no part can anything be transposed without confusion to the rest and to the whole universe.

~ COPERNICUS, *De Revolutionibus*

P.A.M. Dirac is said to have usually opened his course in quantum mechanics with the remark: "The existence of an external world is assumed. That is all the metaphysics you will need for this course."

~ STILLMAN DRAKE, *Galileo: Pioneer Scientist*

Writers who know nothing of the true poetry of modern science have supposed that the perception of the sublime is born of ignorance, and that to admire, it is necessary not to know. This is assuredly a strange error.

~ FLAMMARION AND GORE, *Popular Astronomy*

The enchanting charms of this sublime science reveal themselves only to those who have the courage to go deeply into it. But when a woman, who because of her sex and our prejudices encounters infinitely more obstacles than a man in familiarizing herself with complicated problems, succeeds nevertheless in surmounting these obstacles and penetrating the most obscure parts of them, without doubt she must have the noblest courage, quite extraordinary talents and superior genius.

~ GAUSS, *Letter to Sophie Germain (1807)*

One of the principal objects of theoretical research is to find the point of view from which the subject appears in the greatest simplicity.

~ JOSIAH W. GIBBS

Although mathematics is clearly of huge importance in the development of physics, one of the greatest achievements in all of science Darwin's achievement in*On the Origin of Species*makes no real use of mathematics.

~ PETER GODFREY-SMITH, *Theory and Reality*

For Popper, an inspiring example of genuine science was the work of Einstein. Examples of pseudo science were Freudian psychology and Marxist views about society and history.

~ PETER GODFREY-SMITH, *Theory and Reality*

And crucially, for Popper it is never possible to confirm or establish a theory by showing its agreement with observations.*Confirmation is a myth.*The only thing that an observational test can do is to show that a theory is false.

~ PETER GODFREY-SMITH, *Theory and Reality*

The humanities had gone to hell, and now they were trying to wreck science as well, via endless relativist bleating that science is "just another approach to knowledge with no special status."

~ PETER GODFREY-SMITH, *Theory and Reality*

Mathematics is rich, even dense, with interconnections, but it exhibits no unity.

~ IVOR GRATTAN-GUINESS, *The Norton History of the Mathematical Sciences*

In addition, the teaching of theories from axioms, or some close imitation of them such as the basic laws of algebra, is usually an educational disaster; for whatever the gains in rigour, they are achieved at the expense of a heavy loss of heuristic understanding and intuitive reasoning the very things that lie at the heart of real educational theory.

~ IVOR GRATTAN-GUINESS, *The Norton History of the Mathematical Sciences*

...set theory and topology were extraordinarily late arrivals on the mathematical scene.

~ IVOR GRATTAN-GUINESS, *The Norton History of the Mathematical Sciences*

Neither Newton nor Leibniz married, nor did they seem to be "interested".

~ IVOR GRATTAN-GUINESS, *The Norton History of the Mathematical Sciences*

It is in this gesture of "going beyond", to be something in oneself rather than the pawn of a consensus, the refusal to stay within a rigid circle that others have drawn around one it is in this solitary act that one finds true creativity. All other things follow as a matter of course.

~ ALEXANDER GROTHENDIECK, *Récoltes et semailles*

*Reduction ad absurdum,*which Euclid loved so much, is one of a mathematician's finest weapons. It is a far finer gambit than any chess gambit; a chess player may offer the sacrifice of a pawn or even a piece, but a mathematician offers the game.

~ G. H. HARDY

..what we observe in nature is not nature in itself but nature exposed to our method of questioning.

~ WERNER HEISENBERG, *Physics and Philosophy*

The idea that electromagnetic waves could be a reality in themselves, independent of any bodies, did at that time not occur to the physicists.

~ WERNER HEISENBERG, *Physics and Philosophy*

The electrodynamics of moving bodies can be derived at once from the principle of relativity.

~ WERNER HEISENBERG, *Physics and Philosophy*

One of the truly universal concepts that runs through almost every phase of mathematics is that of a *function,* a *mapping* from one set to another. One could safely say that there is no part of mathematics where the notion does not arise or play a central role.

~ I. N. HERSTEIN, *Abstract Algebra*

I do not see that the sex of a candidate is an argument against her [Emmy Noether's] admission as *Privatdocent.* After all, we are a university, not a bathing establishment.

~ DAVID HILBERT

One pattern of 0s and 1s is converted into another, and that is what computation is.

~ GEORGE JOHNSON, *A Shortcut Through Time*

Experience, not books, is what leads to understanding.

~ CARL JUNG, *Psychology and Alchemy*

Civilization advances by extending the number of important operations which we can perform without thinking about them.

~ CARL JUNG, *Whitehead, Alfred North*

The practical use of mathematics has an importance which it would be foolish to deny. Without mathematics such amenities of modern civilization such as elevators, automatic cigarette vending machines and atomic bombs could not have been developed.

~ KERSHNER AND WILCOX, *The Anatomy of Mathematics*

The need to recognize the existence of an undefined basis [assumptions] cannot be overemphasized.

~ KERSHNER AND WILCOX, *The Anatomy of Mathematics*

It is regrettable that mathematical words should have other meanings as well, since a nonmathematical meaning tends to influence understanding of the mathematical meaning. Thus,*set, function, relation,*and*operation* have mathematical meanings that are entirely, or almost entirely, divorced from their everyday meanings. The reader should not expect *real numbers*to be any more real, or any less imaginary, than*imaginary* numbers. There is nothing even remotely irrational about*irrational numbers.*These are all equally straightforward mathematical entities which happen to be unfortunately named.

~ KERSHNER AND WILCOX, *The Anatomy of Mathematics*

By a number, then, until further notice, we shall mean one of a particular collection of grunts (differing with the language) associated with the ritual of counting.

~ KERSHNER AND WILCOX, *The Anatomy of Mathematics*

In fact mathematics had developed illogically. Its illogical development contained not only false proofs, slips in reasoning, and inadvertent mistakes which with more care could have been avoided. Such blunders there were aplenty. The illogical development also involved inadequate understanding of concepts, a failure to recognize all the principles of logic required, and an inadequate rigor of proof; that is, intuition, physical arguments, and appeal to geometrical diagrams had taken the place of logical arguments.

~ MORRIS KLINE, *The Loss of Certainty*

Gödel's theorems produced a debacle. It is now apparent that the concept of a universally accepted, infallible body of reasoning the majestic mathematics of 1800 and the pride of man is a grand illusion.

~ MORRIS KLINE, *The Loss of Certainty*

God created the integers. All else is the work of man.

~ LEOPOLD KRONEKER

There is no way of proving that a conceptual scheme is final.

~ THOMAS S. KUHN, *The Copernican Revolution*

Men who believed that their terrestrial home was only a planet circulating blindly about one of an infinity of stars evaluated their place in the cosmic scheme quite differently than had their predecessors who saw the earth as the unique and focal center of God's creation.

~ THOMAS S. KUHN, *The Copernican Revolution*

The initial battle between Copernicus and the astronomers of antiquity was fought over technical minutiae.

~ THOMAS S. KUHN, *The Copernican Revolution*

Copernicus himself believed in neither [infinite space nor the possibility of a vacuum]. As we shall see, he tried to preserve most of the central features of Aristotelian and Ptolemaic cosmology.

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Copernicus belonged to the minority group of Renaissance astronomers who did not cast horoscopes.

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*De Revolutionibus*is a relatively staid, sober, and unrevolutionary work.

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Copernicus' system is neither simpler nor more accurate than Ptolemy's.

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The significance of*De Revolutionibus*lies, then, less than in what it says itself than in what it caused others to say. The book gave rise to a revolution that it had scarcely enunciated.

~ THOMAS S. KUHN, *The Copernican Revolution*

The Copernican Revolution, as we know it, is scarcely to be found in the *De Revolutionibus*and that is the second essential incongruity of the text.

~ THOMAS S. KUHN, *The Copernican Revolution*

During the century and a half following Galileo's death in 1642, a belief in the earth centered universe was gradually transformed from an essential sign of sanity to an index, first, of inflexible conservatism, then of excessive parochialism, and finally of complete fanaticism.

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To be accepted as a paradigm, a thing must seem better than its competitors, but it need not, and in fact never does, explain all the facts with which it can be confronted.

~ THOMAS S. KUHN, *The Structure of Scientific Revolutions*

The temptation to write history backward is both omnipresent and perennial.

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The infinite, whether the infinitely large or the infinitely small, seems to carry disaster in its wake.

~ J. W. LASLEY, *The Revolt Against Aristotle*

Ordinary languages, though mostly helpful for the inferences of thought, are yet subject to countless ambiguities and cannot do the task of a calculus, which is to expose mistakes in inference. This remarkable advantage is afforded up to date only by the symbols of arithmeticians and algebraists, for whom inference consists only in the use of characters, and a mistake in thought and in the calculus is identical.

~ LEIBNIZ

Above all, the ambiguities underlying the explanations of the calculus had to be eliminated. This book chronicles the work of these men and their heirs and, in so doing, traces the way from the calculus to point set topology.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The vague and inconsistent notions of limit which accompanied the expositions of Newton and Leibniz created difficulties which they and their followers were unable to resolve. For the next two centuries the attention of the leading mathematicians was directed towards this problem of nearnes; its ultimate satisfactory solution provided both a basis and a motivation for the development of point set topology.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The success of mathematicians in geometrically conceptualizng finite processes naturally suggested an extension of this technique to include infinite processes. The recurrent failure of this sort of conceptualization had many consequences.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

"In a word then, there are mathematicians who, in their kingdom of abstraction, assume indivisible substances which are without parts, without length, and without width, but not, however, the physicists for whom, working in the field of matter no such thing is permitted." [Gasendi, 1658]

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The difficulty with the differentiation process is that there is no smooth conceptual transition from the geometrical representation of the difference quotient to that of the derivative.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

...in the pre-Weierstrassian period, the concepts of the infinitesimal were necessarily geometric.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The importance of convergence criteria in infinite series investigations was slow to be recognized.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The philosophical position maintained by Augustin-Louis Cauchy enabled him to produce the first successes in this direction [limits]. "But it would be a serious error to think that one can only find certainty in geometrical considerations or in the testimony of the senses." [1821]

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Instead of banishing intuition from mathematics for all time, [Cauchy] had succeeded only in driving it to a deeper level of consciousness.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The analysis of the vibrating string gave a first extension to the notion of function and inaugurated researches concerned with sets of points.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The view that geometric form transcends analytical expression was necessarily replaced by the belief that analytic expression can represent every curve that can be drawn.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

With Riemann's example of an integrable function that possesses an infinite number of discontinuities between any two limits the function concept had achieved a generality in excess of that enjoyed by graphs or traced curves. The full impact of this generality was not, however, recognized until an even more startling example due to Weierstrass was presented some years later.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The concept that the senses are adequate determinants of mathematical truth was slow to be replaced. Yet the total rejection of this sensory role was prerequisite to the development of modern analysis.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The idea that a function can be based upon the more primitive notion of set had its inception in Dirichlet's work [~1830].

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The disquieting thesis that pictorial representation did not have a greater generality than analytic construction was neither rapidly nor happily accepted when presented by Fourier. The suggestion that the inclusion actually went the other way so disturbed the equilibrium of the structure of mathematics (and mathematicians) that more than four decades elapsed from discovery (Bolzano*circa*1830) to print (du Bois-Reymond's exposition of Weierstrauss' function, 1874). The same sense of security which caused the early analysts to refer the limit concept to motion operated almost two centuries later in the reluctance to abandon geometric referents.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Overcoming the reliance on sensory perception was a necessary step in the evolution of topology.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Indeed, arithmetization demonstrated the thesis that mathematicas has an existence independent of the perceptual world and that, insofar as mathematics,*qua*mathematics, is concerned, its intersection with reality in totally irrelevant. This concept, a major characteristic of modern mathematics, was indispensable to the development of the notion of a topological space.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Set theory was slow to mature for two reasons. In the first place, as has been seen, considerations of sets of points were originally subsidiary to some other investigation. Another reason for the delayed appearance of a theory of sets relates to the nature of the set concept itself. Many of the notions employed in mathematics have a basis in intuition, area and dimension being examples. Such examples appear to have a visual referent; a certain status is thereby given these intuitive notions. The associated numbers evoke a certain image of the concept, e.g. an area of five square units, a dimension of two. The set concept is different. A finite point set is rather uninteresting. If the word*finite*is replaced by *infinite,*then sensory contact is impossible.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Hausdorff selected the neighbourhood concept as fundamental [instead of distance or limit].

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Attempts to establish a general proof of absolute consistency (proof theory) have been abandoned as a result of a theorem by Gödel who showed that there are, in any sufficiently strong system, undecidable problems, one of which is that of the consistency of the system [1931].

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

"What may be called the Euclid–Klein concept of geometry, that is the study of equivalence under a given group of transformations, was replaced by the idea that a space has an intrinsic structure, consisting of a set of relations which may be, but in general are not, defined in terms of a transformation group." [Whitehead]

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Paradoxes of the infinite, discussed by the Greeks, became urgent problems with the codification of calculus.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The movement towards abstraction and generalization in mathematics is a measure of progress towards the notion of a topological space.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

But the historical development of mathematics appears to be characterized by a reaching ot towards both more complex and more elemental notions from starting points that are determined by man's contacts with the physical world.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

Cantor saw that a theory of sets had implications for all branches of mathematics and, as had been anticipated by Dirichlet, that the set concept was more fundamental than that of function.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

In his 1906 thesis [Frechet] considered first an arbitrary set of elements *E*and an operation which maps*E*into some numerically determined value.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

The appearance in 1914 of Hausdorff's*Grundzüge der Mengenlehre*marks the appearance of point set topology as a separate discipline. A*space* became merely a set of points and a set of relations involving these points, anda geometry was simply the theorems concerning the space. Hausdorff began his development of topology with a small non–categorical set of neighborhood axioms. After deriving many of the properties of the general space thus defined he progressively introduce new axioms, ultimately deducing metric spaces and finally, particular Euclidean spaces.

~ JEROME H. MANHEIM, *The Genesis of Point Set Topology*

When chance is involved, people's thought processes are often seriously flawed.

~ LEONARD MLODINOW, *The Drunkard's Walk*

We habitually underestimate the effect of randomness.

~ LEONARD MLODINOW, *The Drunkard's Walk*

Complete chaos is ironically a kind of perfection.

~ LEONARD MLODINOW, *The Drunkard's Walk*

Avoiding the illusion of meaning in random patterns is a difficult task.

~ LEONARD MLODINOW, *The Drunkard's Walk*

...we should learn to spend as much time looking for evidence that we are wrong as we spend searching for reasons we are correct.

~ LEONARD MLODINOW, *The Drunkard's Walk*

Galileo was not one of the great mathematical astronomers. He remained loyal to the circles used by Copernicus in planetary theory, and even in his*Dialogue Concerning the Two Chief World Systems*he made no reference to the elliptical orbits introduced by Kepler, although he certainly knew of Kepler's work.

~ JOHN NORTH, *Cosmos*

...when God created [the heavens], He impressed them with a certain quality and force of motion, just as He impressed terrestrial things with weight; it is just the same as a man building a clock and leaving it to run itself. [Written ca. 1360]

~ ORESME

In the very first sentence ever written on quantum mechanics, the abstract of Heisenberg's July 1925 paper, it was stated that "quantum mechanics should be founded exclusively upon relationships between quantities that are in principle observable."

~ ABRAHAM PAIS, *Niels Bohr*

At the end of [Heiseberg's] first lecture [at age 24] one student reportedly said that one wouldn't have believed he was so clever since he looked like a bright carpenter's apprentice just returned from technical school.

~ ABRAHAM PAIS, *Niels Bohr*

Planck most probably first wrote down his law on the evening of Sunday 7 October 1900, as the result of new experimental information he received that day.

~ ABRAHAM PAIS, *Niels Bohr*

Anything is easy if you can assimilate it to your collection of models. If you can't, anything can be painfully difficult.

~ SEYMOUR PAPERT, *Mindstorms*

...using the computer as an "object to think with".

~ SEYMOUR PAPERT, *Mindstorms*

Stated most succinctly, my conjecture is that the computer can concretize (and personalize) the formal.

~ SEYMOUR PAPERT, *Mindstorms*

But making good choices is not always easy, in part because past choices can often haunt us. I have called this phenomenon the QWERTY phenomenon.

~ SEYMOUR PAPERT, *Mindstorms*

I see "school math" as a social construction, a kind of QWERTY.

~ SEYMOUR PAPERT, *Mindstorms*

One might even say that computer science is wrongly so called. Most of it is not the science of computers, but the science of descriptions and descriptive languages.

~ SEYMOUR PAPERT, *Mindstorms*

First, relate what is new and to be learned to something you already know. Second, take what is new and make it your own: make something new with it, play with it, build with it.

~ SEYMOUR PAPERT, *Mindstorms*

For if you let two weights fall from the same height, one of them being many times heavier than the other, you will see that the times required for the motion are not in proportion dependent on the ratio of their weights, but that the actual difference in time is very small. [6th cent.]

~ PHILOPONAS THE BYZANTINE

...the woof and warp of all thought and research is symbols, and the life of thought and science is the life inherent in symbols; so it is wrong to say that a good language is important to good thought, merely; for it is the essence of it.

~ CHARLES PIERCE

The statement that the universe has to have had this history in order for galaxies, stars, and ourselves to exist is verging on the tautological and hardly merits the title*anthropic principle*which some authors give it.

~ M. ROWAN-ROBINSON, *Cosmology*

Our achievements rest on the accomplishments of 40,000 generations of our human predecessors, all but a tiny fraction of which are nameless and forgotten.

~ CARL SAGAN, *Cosmos*

...these are some of the things that hydrogen atoms do, given fifteen billion years of cosmic evolution.

~ CARL SAGAN, *Cosmos*

The framework that modern natural philosophers preferred to Aristotelian teleology was one that explicitly modelled nature on the characteristics of a*machine.*

~ STEVEN SHAPIN, *The Scientific Revolution*

The machine metaphor might, then, be a vehicle for "taking the wonder out" of our understanding of nature or, as the sociologist Max Weber put it in the early twentieth century, for the "disenchantment of the world".

~ STEVEN SHAPIN, *The Scientific Revolution*

According to Hilbert, the most vulnerable point in the fortress of mathematics was the infinite.

~ STEVE SIMPSON, *Partial Realizations of Hilbert's Program*

It therefore behooves the historian of science to be very charitable, very forbearing, very humble, in his judgements and presentations of those who have gone before him. He needs to remember that he is dealing with the work of erring and imperfect human beings.

~ C. SINGER, *A History of Scientific Ideas*

In the Middle Ages languages were learned by speaking and not from grammars.

~ C. SINGER, *A History of Scientific Ideas*

It appeared about 1240 and was still being reprinted in the seventeenth century. It is one of the puzzles of history that this great improvement [Hindu Arabic numerals], represented by the "Arabic" as against the Latin system, took three centuries to gain general acceptance.

~ C. SINGER, *A History of Scientific Ideas*

The humanists discovered the literary works of antiquity. In them they became absorbed to the exclusion of all else. Their eagerness passed into a literary vogue, and cast the blight of a purely literary education on the modern world.

~ C. SINGER, *A History of Scientific Ideas*

The most cherished goal in physics, as in bad romance novels, is unification.

~ LEE SMOLIN, *The Trouble With Physics*

To be fair, we have made two experimental discoveries in the past few decades: that neutrinos have mass and that the universe is dominated by a mysterious dark energy that seems to be accelerating its expansion.

~ LEE SMOLIN, *The Trouble With Physics*

On the other hand, if string theorists are wrong, they can't be just a little wrong. If the new dimensions and symmetries do not exist, then we will count string theorists among science's greatest failures, like those who continued to work on Ptolemaic epicycles while Kepler and Galileo forged ahead. Theirs will be a cautionary tale of how not to do science, how not to let theoretical conjecture get so far beyond the limits of what can rationally be argued that one starts engaging in fantasy.

~ LEE SMOLIN, *The Trouble With Physics*

Some people need to think through everything very carefully, and this takes time, as they get easily confused. It's not hard to feel superior to such people, until you remember that Einstein was one of them.

~ LEE SMOLIN, *The Trouble With Physics*

What is remarkable to me is the number of distinguished scientists who seem unable to accept the possibility either that string theory or the hypothesis of a random multiverse is wrong.

~ LEE SMOLIN, *The Trouble With Physics*

...it is not good for any field if any one person's views are taken too authoritatively. There is no scientist, not even Newton or Einstein, who was not wrong on a substantial number of issues they had strong views about.

~ LEE SMOLIN, *The Trouble With Physics*

If your ideas are right and you fight for them, you'll accomplish something. No one but you can develop your ideas, and no one but you will fight for them.

~ LEE SMOLIN, *The Trouble With Physics*

But what is equally important, and sobering, is how often we fool ourselves. And we fool ourselves not only individually but en masse. The tendency of a group of human beings to quickly come to believe something that its individual members will later see as obviously false is truly amazing.

~ LEE SMOLIN, *The Trouble With Physics*

Feyerabend insisted that scientists should never agree, unless they are forced to.

~ LEE SMOLIN, *The Trouble With Physics*

If a highly disciplined subject like physics is vulnerable to the symptoms of groupthink, what may be happening in other, less rigorous areas?

~ LEE SMOLIN, *The Trouble With Physics*

[String theorists] seem to feel that appeal to consensus within their community is equivalent to rational argument.

~ LEE SMOLIN, *The Trouble With Physics*

Good ideas are not taken seriously enough when they come from people of low status in the academic world; conversely, the ideas of high status people are often taken too seriously.

~ LEE SMOLIN, *The Trouble With Physics*

The change over from a developmental to a static conception of matter was as profound as the change from a geocentric to a heliocentric astronomy, and its effects were as far reaching.

~ TOULMIN AND GOODFIELD, *The Architecture of Matter*

Yet ideas cannot be burned at the stake as heretics can.

~ TOULMIN AND GOODFIELD, *The Fabric of the Heavens*

For Galileo still thought of "inertial motion" as naturally circular, whether in the case of a ship moving over the Earth's surface, or in that of a satellite circling around the planet Jupiter.

~ TOULMIN AND GOODFIELD, *The Fabric of the Heavens*

It says much for Newton's mathematical facility that he engaged in the whole enterprise as though it were an intellectual pastime, while believing all the time that the questions of true importance lay in other, religious directions.

~ TOULMIN AND GOODFIELD, *The Fabric of the Heavens*

Leibniz would have said that the term "field" was only a way of making Newton's miracles computable.

~ TOULMIN AND GOODFIELD, *The Fabric of the Heavens*

Indeed, it was Descartes and not Galileo who stated the principle of inertia in the form we accept today, insisting on the rectilinear character of inertial motion.

~ RICHARD WESTFALL, *Never at Rest*

For them, Aristotelian philosophy was dead beyond resurrection. In its place stood a new philosophy for which the machine, not the organism, was the dominant analogy.

~ RICHARD WESTFALL, *Never at Rest*

...when Pierre Fermat derived the law [of refraction] from the principle of least time, mechanical philosophers refused to have truck with what they considered occult notions.

~ RICHARD WESTFALL, *Never at Rest*

Nevertheless, I have undertaken to write a biography of Newton, and my personal preferences cannot make more than a million words he wrote in the study of alchemy disappear.

~ RICHARD WESTFALL, *Never at Rest*

By 1661, the official curriculum of Cambridge, prescribed by statute nearly a century before, was in an advanced state of decomposition.

~ RICHARD WESTFALL, *Never at Rest*

What he thought on, he thought on continually, which is to say exclusively, or nearly exclusively. What seized his attention in 1664, to the virtual exclusion of everything else, was mathematics.

~ RICHARD WESTFALL, *Never at Rest*

Cambridge was fast approaching the status of an intellectual wasteland.

~ RICHARD WESTFALL, *Never at Rest*

The first point is that the enormous usefulness of mathematics in the natrual sciences is something bordering on the mysterious and that there is no rational explanation for it.

~ EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

Furthermore, whereas it is unquestionably true that the concepts of elementary mathematics and particularly elementary geometry were formulated to describe entities which are directly suggested by the actual world, the same does not seem to be true of the more advanced concepts, in particular the concepts which play such an important role in physics.

~ EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

Without invariance principles physics would not be possible.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

The preceding discussion is inteded to remind, first, that it is not at all natural that "laws of nature" exist, much less that man is able to discover them.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

...the construction of machines, the functioning of which he can foresee, constitutes the most spectactular accomplishment of the physicist.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

The statement that the laws of nature are written in the language of mathematics was properly made three hundred years ago; it is now more true than ever before.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

However, it is important to point out that the mathematical formulation of the physicist's often crude experience leads in an uncanny number of cases to an amazingly accurate description of a large class of phenomena.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

The miracle occurred only when matrix mechanics, or a mathematically equivalent theory, was applied to problems for which Heisenberg's calculating rules were meaningless. Surely in this case we "got something out" of the equations that we did not put in.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

The event of coincidence, that is in ultimate analysis of collision, is the primitive event in the theory of relativity and defines a point in space-time, or at least would define a point if the colliding particles were infinitely small.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

Fundamentally, we do not know why our theories work so well.

EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

The miralce of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve.

~ EUGENE P. WIGNER, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*

Quadrance and spread are quadratic quantities, while distance and angle are almost, but not quite, linear ones. The quadratic view is more general and powerful. At some level, this is known by many mathematicians. When this insight is put firmly into practice, as it is here, a new foundation for mathematics and mathematics education arises which simplifies Euclidean and non–Euclidean geometries, changes our understanding of algebraic geometry, and often simplifies difficult practical problems.

~ N. WILDBERGER, *Divine Proportions*

In fact rational trigonometry is so elementary that almost all calculations may be done by hand. Tables or calculators are not necessary, although the latter certainly speed up computations. It is a shame that this theory was not discovered earlier, since accurate tables were for many centuries not widely available.

~ N. WILDBERGER, *Divine Proportions*